WORKED SOLUTION

STEP I 1989, Question 14

Probability and Statistics • Conditional Distributions

STEP I 1989, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q14

The key idea

Everything rests on one set of probabilities: the chance that a seed lands in each league-wide strip. A seed grows into a tree exactly when it lands in the parkland or the farmland, so the number of trees is binomial. The distance Y at which a tree grows is just L *conditioned* on landing on land. Its density is f cut down to the land strips and scaled up so that the total area is 1. The final part is a hypothesis test: is 23 trees on the farmland surprisingly many for a binomial count?

Solution

The strip probabilities

The cumulative distribution function of L is, for $0 \le x \le 4$,

$$\mathrm{F}(x) = \int_0^x \left(\tfrac{1}{2} - \tfrac{1}{8}t\right)\mathrm{d}t = \frac{x}{2} - \frac{x^2}{16},$$

so $\mathrm{F}(1) = \frac{7}{16}$, $\mathrm{F}(2) = \frac{3}{4}$, $\mathrm{F}(3) = \frac{15}{16}$ and $\mathrm{F}(4) = 1$. Hence

$$\mathrm{P}(\text{parkland}) = \mathrm{P}(0 \le L \le 1) = \tfrac{7}{16}, \qquad \mathrm{P}(\text{lake}) = \tfrac{3}{4} - \tfrac{7}{16} = \tfrac{5}{16},$$
$$\mathrm{P}(\text{farmland}) = \mathrm{P}(2 \le L \le 3) = \tfrac{15}{16} - \tfrac{3}{4} = \tfrac{3}{16}, \qquad \mathrm{P}(\text{sea}) = \tfrac{1}{16}.$$

(The boundaries have probability zero, since L is continuous, so it doesn't matter which strip they belong to.) A seed grows into a tree with probability

$$p = \tfrac{7}{16} + \tfrac{3}{16} = \tfrac{5}{8}.$$

Let T be the number of trees in a year. Since the 96 seeds are blown independently, each growing with probability $\frac{5}{8}$, we have $T \sim \mathrm{B}(96, \frac{5}{8})$, and the mean number of new fever trees each year is

$$\mathrm{E}(T) = 96 \times \tfrac{5}{8} = 60.$$

Part (i)

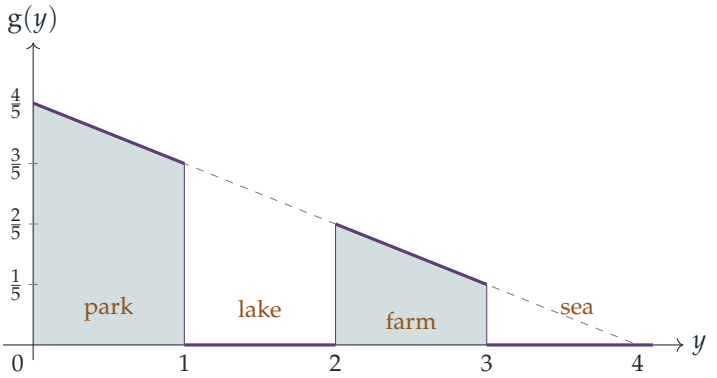
Y has the distribution of L given that the seed lands on land. Write $A = [0,1] \cup [2,3]$ for the land. For $y \in A$,

$$\mathrm{P}(y \le Y \le y + \delta y) = \frac{\mathrm{P}(y \le L \le y + \delta y)}{\mathrm{P}(L \in A)} \approx \frac{\mathrm{f}(y)\,\delta y}{5/8},$$

and Y never takes a value outside A. So the density of Y is

$$\mathrm{g}(y) = \begin{cases} \frac{8}{5}\left(\frac{1}{2} - \frac{1}{8}y\right) = \dfrac{4-y}{5} & \text{for } 0 \le y \le 1 \text{ or } 2 \le y \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

As a check, its area is $\frac{8}{5}\left(\frac{7}{16} + \frac{3}{16}\right) = 1$. The graph is two pieces of the same falling line: from $\frac{4}{5}$ down to $\frac{3}{5}$ over the parkland, nothing over the lake, then from $\frac{2}{5}$ down to $\frac{1}{5}$ over the farmland.



The dashed line is $\frac{8}{5}\mathrm{f}$, shown for comparison; the density of Y follows it only over land.

For the mean, $\int y(4-y)\,\mathrm{d}y = 2y^2 - \frac{1}{3}y^3$, so

$$\begin{aligned} \mathrm{E}(Y) &= \frac{1}{5}\Big[2y^2 - \tfrac{1}{3}y^3\Big]_0^1 + \frac{1}{5}\Big[2y^2 - \tfrac{1}{3}y^3\Big]_2^3 \\ &= \frac{1}{5}\left(\frac{5}{3}\right) + \frac{1}{5}\left(9 - \frac{16}{3}\right) = \frac{1}{5}\cdot\frac{5}{3} + \frac{1}{5}\cdot\frac{11}{3} = \frac{16}{15}. \end{aligned}$$

So a new tree grows, on average, $\frac{16}{15}$ leagues south of the castle. That's just past the parkland, in the lake, which is fine: a mean need not be a value the variable can take.

Part (ii)

Let X be the number of trees that grow in the farmland in a year. If the tradition is followed, 96 seeds are released, and each lands in the farmland independently with probability $\frac{3}{16}$, so

$$X \sim \mathrm{B}\left(96, \tfrac{3}{16}\right), \qquad \mathrm{E}(X) = 18, \qquad \mathrm{Var}(X) = 96 \cdot \tfrac{3}{16} \cdot \tfrac{13}{16} = \tfrac{117}{8} \approx 14.6,$$

giving a standard deviation of about 3.82.

Ask yourself What exactly are we testing, and in which direction?

The null hypothesis is that the tradition was followed: 96 seeds, so $X \sim \mathrm{B}\left(96, \frac{3}{16}\right)$. The natural suspicion behind *more* farmland trees than usual is that more than 96 seeds were released, so we use a one-tailed test and ask how likely 23 *or more* is under the null hypothesis.

Since $np = 18$ and $n(1-p) = 78$ are both large, we use the normal approximation $X \approx \mathrm{N}(18,\, 14.625)$ with a continuity correction:

$$\mathrm{P}(X \geq 23) \approx \mathrm{P}\left(Z \geq \frac{22.5 - 18}{3.824}\right) = \mathrm{P}(Z \geq 1.18) \approx 0.12.$$

(The exact binomial value is 0.121.) So under normal conditions, 23 or more farmland trees happens about one year in eight. That's well short of significance at the 5% level (or even 10%). The observation is quite consistent with the tradition having been followed, and the king is *not* justified in his suspicions on this evidence alone.

For comparison, even at the 5% level the count would have to be at least 25 (exact $\mathrm{P}(X \geq 25) \approx 0.049$) before the king would have reasonable grounds. If he also had the parkland count, he could test the total number of trees, $T \sim \mathrm{B}\left(96, \frac{5}{8}\right)$, which would be much better evidence about how many seeds were released.

Checks

- The four strip probabilities $\frac{7}{16}$, $\frac{5}{16}$, $\frac{3}{16}$, $\frac{1}{16}$ add to 1, and the density of Y has total area 1.
- A midpoint-rule integration gives $\mathrm{E}(Y) = 1.06667$, matching $\frac{16}{15}$. A simulation of 400 000 seeds (using $L = 4 - 4\sqrt{1-U}$ with U uniform on $[0,1]$) found 62.5% landing on land and a mean landing distance of 1.069 among them.
- Summing the binomial probabilities exactly gives $\mathrm{P}(X \geq 23) = 0.1213$, close to the normal estimate of 0.120.

Common slips

- Using f itself as the density of Y. It has area only $\frac{5}{8}$ over the land, so it must be divided by $\frac{5}{8}$.
- Computing $\mathrm{E}(L\,;\, L \in A) = \int_A y\,\mathrm{f}(y)\,\mathrm{d}y = \frac{2}{3}$ and forgetting to divide by $\mathrm{P}(L \in A)$.
- Testing $\mathrm{P}(X = 23)$ rather than the tail $\mathrm{P}(X \geq 23)$. Any single value of X is fairly unlikely.

Beyond STEP

Part (i) is an example of a *conditional distribution*: restricting a random variable to an event and renormalising. The same operation underlies Bayes' theorem for densities and the truncated distributions used throughout statistics. Part (ii) is a first taste of hypothesis testing and p-values, which first-year courses make precise with the Neyman–Pearson framework. Blitzstein and Hwang, *Introduction to Probability*, treats conditional distributions with plenty of worked stories like this one.

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